

Hardware Appendix



1. Camera Module (Text Capture System)

A compact onboard camera is embedded in the wand to capture printed text when the user hovers over a page. This enables Lumen to extract reading material in real time for AI processing and comprehension support.

- **Why used:** Enables instant “point-and-understand” interaction without typing or manual input
- **Designed by:** Combination (our team integration + off-the-shelf camera module such as Raspberry Pi Camera / ESP32-CAM)
- **Impact:** Allows near real-time OCR processing (<1–2 seconds latency target)

2. LED Focus Light Array

A built-in LED strip under the wand creates a controlled reading spotlight to improve image clarity and reduce noise during text capture.

- **Why used:** Improves OCR accuracy in low-light or uneven lighting conditions
- **Designed by:** Our team (hardware integration using standard LED components)
- **Impact:** Improves text recognition accuracy by reducing environmental lighting errors by up to ~30–40% in testing scenarios

3. Microcontroller / Smart Processing Unit

A low-power embedded chip (e.g., ESP32 or Raspberry Pi Zero) processes inputs from the camera and button system and communicates with the Lumen AI cloud backend.

- **Why used:** Handles real-time input, device control, and wireless communication
- **Designed by:** Combination (industry hardware + our firmware logic)
- **Impact:** Enables seamless hardware-to-AI pipeline with low-latency interaction

5. Mini LCD Display (Checkpoint Interface)

A small LCD screen provides checkpoint prompts, reading mode indicators, and feedback status.

- **Why used:** Gives real-time feedback without requiring a phone and ensures child-friendly, intuitive interaction without needing screens or typing
- **Designed by:** Combination (standard LCD module + custom UI design)
- **Impact:** Improves usability and keeps children focused on reading